JIPING WANG

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SUMMARY

Ph.D. Candidate in Statistics with industry experience in data science through internships at Genentech and Ocular Therapeutix. Experienced in statistical modeling, machine learning, Python, and SQL, with a strong background in simulation, optimization, and high-dimensional data analysis. Develops rigorous, scalable solutions to complex real-world data science problems, combining statistical methodology with computational implementation.

EDUCATION

Florida State University *Tallahassee, FL*
Ph.D. Candidate in Statistics *(GPA: 3.98/4.00)* *Aug. 2020 – Spring 2027 (Expected)*

Xiamen University *Xiamen, China*
M.S. in Mathematical Statistics *(GPA: 3.71/4.00; Rank: 1/17)* *Sep. 2017 - Jun. 2020*
B.S. in Mathematical Statistics *(GPA: 3.77/4.00; Rank: 2/31)* *Sep. 2013 - Jun. 2017*

PROFESSIONAL EXPERIENCE

Product Development and Data Science Intern May 2025 – Nov 2025
Genentech

- Simulated longitudinal visual field data to augment limited real-world datasets, enabling large-scale evaluation of experimental designs and statistical endpoints.
- Quantified trade-offs across experimental designs and demonstrated that cluster-based designs reduced required sample sizes by 40%, supporting more efficient study planning.
- Developed and evaluated slope-based progression endpoints against established event-based metrics, improving sensitivity to progression and enabling earlier detection.
- Implemented statistical models, including piecewise linear regression, hierarchical Bayesian models, and AGIS, to analyze pointwise visual field measurements and quantify progression patterns.

Biostatistics Intern May 2024 – Aug 2024
Ocular Therapeutix

- Conducted simulation studies in R to evaluate statistical models for longitudinal data under varying treatment-effect scenarios.
- Generated and analyzed simulated longitudinal datasets with repeated measurements, evaluating model performance through hypothesis testing and treatment-effect estimation.
- Applied linear mixed-effects models with treatment, time, treatment-by-time interactions, and baseline covariates to assess differences between experimental groups.

SELECTED PROJECTS

Statistical Analysis of Spatial Transcriptomics Data Nov. 2026 – Present

- Reproduced a 2026 method (CONCISE) for ligand–receptor inference in spatial transcriptomics on Visium, Stereo-seq, and MERFISH data, matching the paper's quality-control counts.
- Ran simulations of type I error and power and benchmarked against SpatialDM, LIANA+, MERINGUE, and Copulacci; CONCISE controlled false positives best, flagging 1 of about 1,500 housekeeping-gene control pairs as significant, against roughly 25–80% for the others.
- Currently extending the method to zero-inflated counts and adding a classification step.

Hierarchical Sure Independence Screening for Multiresolution Classification Aug. 2023 – Present

- Invented HierSIS, which lifts sure independence screening from flat to hierarchically structured multinomial responses by screening separately at each node of the class hierarchy, with sure screening guarantees generalizing Fan & Song (2010).
- Demonstrated higher true-signal recovery than flat SIS and penalized regression competitors in simulation studies across $p \in \{1500, 2000, 3000, 5000\}$ predictors.

· Applied to the Hao et al. (2021) PBMC scRNA-seq dataset (152,244 cells, 27,373 genes, 24 fine cell types nested in 11 coarse classes), improving macro-F1 by 0.02 and balanced error rate by 0.03 over the leading competitor (mrMLR).

Matrix-Response Regression with Low Tubal Rank and Sparsity *(Manuscript)* Jun. 2022 – Dec 2025

· Proposed a matrix-response regression framework integrating low tubal-rank and sparsity constraints for structured high-dimensional data.
· Designed proximal gradient and linearized ADMM algorithms with convergence guarantees; proved non-asymptotic error bounds and variable selection consistency under restricted strong convexity.
· Benchmarked against 6 competing methods across 9 simulation settings, consistently achieving lower estimation error, better rank recovery, and variable selection consistency across sparsity structures.
· Applied to EEG seizure data (19 electrodes, 60 timepoints), pinpointing electrodes P4, C4, and T4 as the strongest seizure biomarkers, consistent with established neurological findings.

Censored Mean Variance Sure Independence Screening for Ultra-high Dimensional Survival Data Mar. 2018 – Oct. 2019

· Developed a model-free feature screening method for ultra-high-dimensional survival data under censoring, with rigorous theoretical guarantees
· Established sure screening and ranking consistency properties under mild regularity conditions, supported by formal proofs.
· Conducted extensive simulation studies demonstrating near-100% true-variable inclusion rates under 20%–40% censoring.
· Applied the method to breast cancer survival data, achieving the strongest overlap with consensus gene sets and the lowest Kolmogorov Smirnov distance and Cramer-von Mises distance for Cox model compared with competing methods.

TEACHING EXPERIENCE

Florida State University Aug. 2021 – May 2026

· Sole Instructor for STA 1013 (Statistics Through Examples) for 3 semesters; designed and taught the full course, combining mathematical derivations with real datasets in R.
· Teaching Assistant for courses covering regression, probability, statistical computing, and functional data analysis.

PUBLICATIONS

Jiping Wang, Xin Zhang. "Matrix-Response Regression with Low Tubal Rank and Sparsity". *Submitted*

Wei Zhong, **Jiping Wang**, Xiaolin Chen. "Censored mean variance sure independence screening for ultrahigh-dimensional survival data". *Computational Statistics & Data Analysis 2021.*

TECHNICAL SKILLS

Programming: Python, SQL, R, MATLAB

Machine Learning: Statistical Learning, Deep Learning, Large Language Models (LLMs), Natural Language Processing (NLP)

Statistics: A/B Testing, Causal Inference, Bayesian Modeling, Missing Data, High-Dimensional Statistics

Optimization: ADMM, Proximal Gradient Methods, Large-Scale Simulation

Frameworks: PyTorch, scikit-learn, NumPy, Pandas, Transformers, HPC/Slurm

HONORS

American Statistical Association Florida Chapter Student Presentation Award *Mar. 2024*
Clara Kibler Davis Scholarship *Oct. 2022*
Graduation with Honors (Cum Laude) *Jun. 2020*
Graduation with Honors (Cum Laude) *Jun. 2017*